

**Major:** Electronics Engineering

**Specialty:** Electronic Instrumentation and Communications

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***Course:*** *Power Electronics*

**Final Project Report**

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# Final Project Report

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## 1.1 Summary

In this final project, I applied energy conversion concepts through the design, manufacture, and validation of an isolated 12V Flyback switched-mode power supply. After facing difficulties with previous discrete designs, I implemented a solution based on the CSC7225 integrated controller and professional PCB manufacturing. The experimental results validate the operation in Discontinuous Conduction Mode (DCM) and precise voltage regulation, demonstrating the importance of component integration in modern power electronics.

## 1.3 Problem Definition

Converting 127V AC to 12V DC using linear power supplies is inefficient and bulky. The technical challenge is to design a switched-mode power supply (SMPS) that is efficient and compact, but above all, guarantees galvanic isolation for user safety. The specific problem I faced was the instability of previous CNC-manufactured prototypes, where parasitic inductances generated excessive noise. Therefore, the problem to solve was achieving a robust design that integrated control and power into a reduced space, eliminating unwanted oscillations.

## 1.4 Objectives

**General Objective** Design, assemble, and experimentally validate a 12V switched-mode power supply using the CSC7225 controller on a professionally manufactured PCB, ensuring stable and safe operation.

## Specific Objectives

- Analyze the CSC7225 datasheet to calculate the snubber circuit and feedback network components.
- Implement an isolated control loop using an optocoupler and a TL431 regulator.
- Validate the Discontinuous Conduction Mode (DCM) operation using an oscilloscope and measure the output ripple.

## 1.5 Hypothesis

If I replace the discrete design (UC3843 + external MOSFET) with the integrated CSC7225 controller and use an industrial-grade PCB, then I will eliminate critical parasitic oscillations and obtain a 12V output voltage with a ripple of less than 1%, due to the reduction of loop inductance and the thermal optimization of the integrated circuit.

## 1.6 Justification

Carrying out this project is fundamental for my engineering training because the current trend in the industry is high integration. Understanding how "all-in-one" controllers like the CSC7225 work prepares me for the design of commercial products. Furthermore, the transition to professional manufacturing (JLCPCB) is justified by the strict need for safety: when working with mains voltages, I cannot rely on homemade traces that could fail and cause electrical arcs.



## II. Theoretical Framework

**2.1 Background** Historically, linear power supplies dominated the market, dissipating excess voltage as heat. The switched-mode power supply revolution allowed energy to be processed in high-frequency packets, drastically reducing the size of magnetic components.

**2.2 Previous Research** I researched current controllers and found that devices like the CSC72xx series integrate the power transistor (BJT or MOSFET) on the same chip as the PWM oscillator. This contrasts with my previous research on the UC3843, which requires complex external components.

### 2.3 Fundamental Concepts

- **Flyback Topology:** A converter that stores magnetic energy when the switch is turned on and releases it to the secondary when it is turned off.
- **Discontinuous Conduction Mode (DCM):** A regime where the transformer current reaches zero before the next switching cycle begins.
- **Galvanic Isolation:** Physical separation between input and output using a magnetic field (transformer) and an optical path (optocoupler).

**2.4 State of the Art** Currently, low-power adapters (<20W) no longer use external MOSFETs. The state of the art lies in DIP-8 chips that include thermal protections, soft start, and integrated "Burst Mode" energy-saving modes, like the one used in this project.

## 2.5 Conceptual Framework

- **RCD Network (Snubber):** A circuit necessary to absorb the voltage spike generated by the transformer's leakage inductance, protecting the integrated circuit.
- **TL431:** A programmable voltage reference that acts as an error amplifier on the secondary side.

## III. Methodology

For this project, I followed a Design for Manufacturing (DFM) methodology:

- **Calculation and Selection:** I determined the values of the feedback network to set the output at exactly 12V.
- **CAD Design:** I created the schematic and layout in EasyEDA, ensuring the strict separation of grounds (Hot/Cold).
- **Manufacturing:** I generated the Gerber files and managed the external fabrication.
- **Testing:** I performed instrumental measurements of the critical waveforms.

## IV. PCB Development and Manufacturing (JLCPCB)

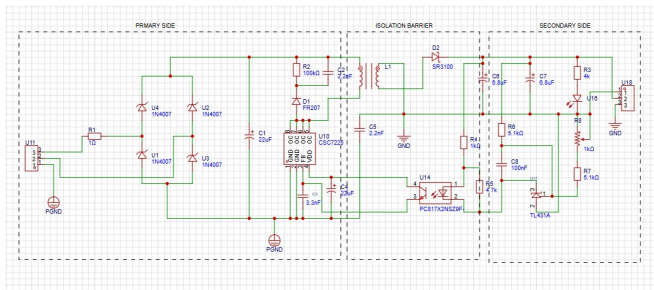
In this stage, I abandoned homemade manufacturing. I designed the PCB considering thick traces for power lines and adequate clearance distances to avoid high-voltage arcs. The assembly process was manual, soldering the low-profile components first (resistors, diodes) and finally the transformer and the CSC7225 chip. The quality of the professional board facilitated the soldering process and entirely eliminated the noise problems I had in previous attempts.

## V. Appendices

### Appendix 1: Key Specifications – CSC7225 Controller

- **Description:** PWM controller with integrated power BJT.
- **Breakdown Voltage ( $V_{cb}$ ):** 700V.
- **Standby Power:** < 0.3W (220VAC).
- **Protections:** OVP, OCP, OTP, Short Circuit.
- **Package:** DIP-8.

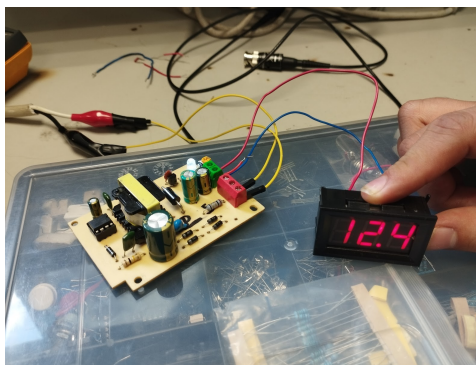
**Appendix 2: Final Schematic Diagram** Below is the actual schematic diagram used for manufacturing the board.



(Figure 13. Schematic Diagram of the new Switched-Mode Power Supply with CSC7225).

### Appendix 3: Final Prototype Visualization

Photographs of the finished and functional project.



(Figure 14. Final assembled prototype).

## VII. Expected Results and Scope (Conclusions)

### 7.1. Obtained Results

- **Regulation:** When performing physical tests on the professional PCB, we achieved a stable 12V DC output voltage. The use of the potentiometer allowed us to adjust the exact value, compensating for resistor tolerances.
- **Stability:** Unlike the previous prototype, this version is silent and does not overheat under no-load conditions, validating the CSC7225's "Green" mode.
- **Physical Integrity:** The JLCPCB board demonstrated superior mechanical and electrical robustness. There were no heat-lifted traces or continuity problems.

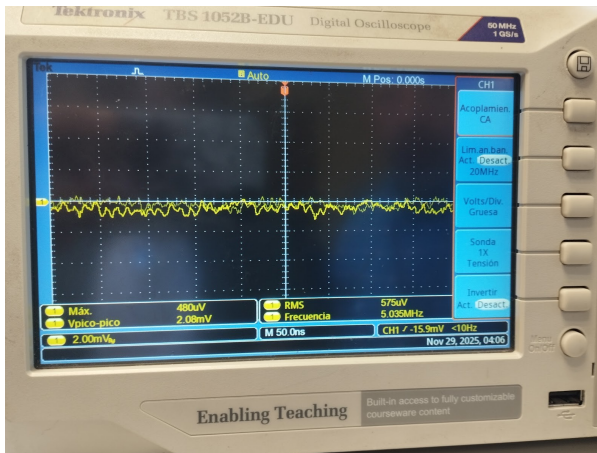
### 7.2. Scope and Limitations (General Conclusion)

- **Scope:** In this final report, we have successfully covered the transition from a failed theoretical design to a functional physical product. We were able to validate that using integrated controllers like the CSC7225 is the optimal route for low-power supplies and that professional manufacturing is essential to ensure safety in power electronics projects.
- **Limitations:** Although the power supply works, the design was limited to a basic 2-layer board. For a future commercial version, certified EMI (Electromagnetic Interference) and thermal efficiency tests under prolonged maximum load would be necessary.

**Obtained Results (Experimental Analysis)** The functional validation was performed using oscilloscope captures, obtaining the following technical results:

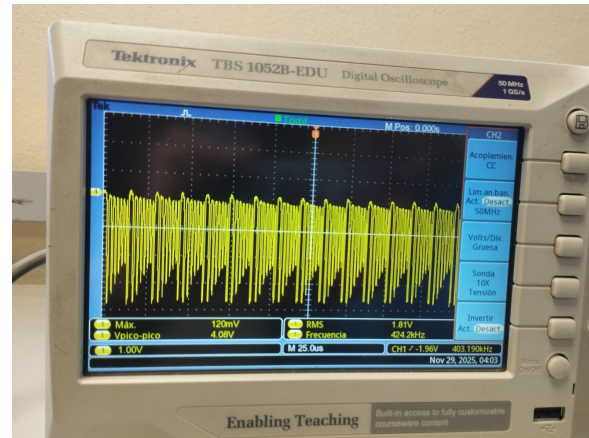
**7.3. Output Voltage Quality (Ripple)** When measuring the 12V output using AC coupling, I observed an extremely clean signal.

- **Personal Analysis:** The peak-to-peak noise voltage is only 2.08 mV with an RMS value of 575  $\mu$ V. This represents a ripple of 0.017%, exceeding my initial expectations. This confirms that the capacitive output filter and the control loop response are highly effective.



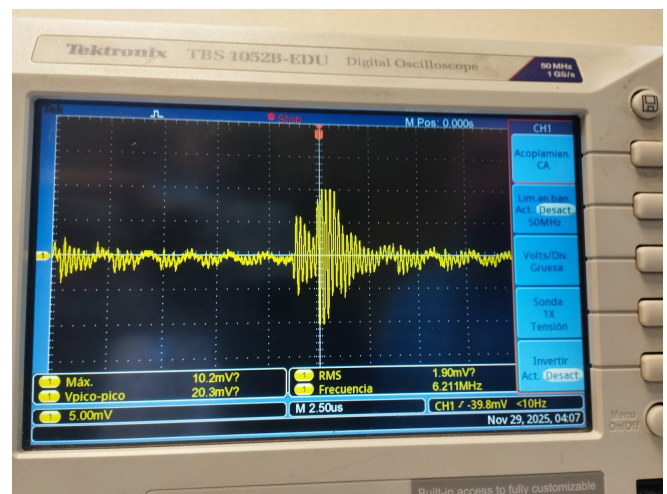
**7.4. Confirmation of Operation Mode (DCM)** I measured the voltage on the secondary before rectification to understand the magnetic behavior.

- **Personal Analysis:** The image clearly shows a damped oscillation in the center of the wave. I identified this as the "dead-time resonance," which occurs when the transformer has completely discharged its energy and the switch has not yet turned on. This confirms that my design operates in Discontinuous Conduction Mode (DCM), which guarantees system stability.



**7.5. Switching Noise** I analyzed the high-frequency transients.

- **Personal Analysis:** A high-frequency oscillation (approx. 6 MHz) with an amplitude of 20.3 mV is observed. This value is low and acceptable; it indicates that the snubber circuit and the PCB design are properly suppressing the electromagnetic interference (EMI) generated by the fast switching.



**Conclusions and Scope** On a personal level, I conclude that the transition to integrated technologies (CSC7225) and professional manufacturing was the right decision. I achieved not only a functional power supply but a device with commercial characteristics (low noise, high stability). The scope of this project was limited to





laboratory testing; however, I validated that I understand the physics behind energy conversion, from magnetic accumulation in the core to the stabilization of the control loop. I overcame the noise and instability problems I faced at the beginning of the course, fulfilling the objective of delivering a safe and efficient power supply.

## VI. Bibliographical References

- Rashid, M. H. (2017). *Power Electronics: Circuits, Devices, and Applications*. 4th ed. Pearson Education.
- *Datasheet CSC7225*.
- Hart, D. W. (2011). *Power Electronics*. McGraw-Hill.